

Igor Pivas

Mechanical Systems Engineer

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📍 Waterloo, ON

EDUCATION

Conestoga College

B. Eng of Mechanical Systems Engineering
Mechanical Engineering Technology Design and Analysis

Cambridge, ON
Jan 2020 – Sept 2022
Sept 2016 – Sept 2019

EXPERIENCE

Systems Designer | Angstrom Engineering | Kitchener, ON

Sept 2022 – Present

- Principal designer for Angstrom Engineering's first commercial Atomic Layer Deposition and Plasma Atomic Layer Deposition systems working from lessons learned from prototype systems
 - Was responsible for all aspects of the design including reactor design, gas delivery and exhaust system, precursor heating system, Peltier precursor cooling system, as well as integration into a multi tool cluster
- Designed over fifteen Physical Vapour Deposition machines which are used for various applications such as perovskite solar panels, optical sensors and quantum computing
- Developed internal tools for determining thin film deposition uniformity in Python which has sped up the design process as well as allowing customers to get an estimate on machine performance
- Eliminated condensation on sub-zero thermal lines through the design of a sealed nitrogen purge housing

SLA Mechanical Engineering Intern | Formlabs | Somerville, MA, USA

Sept 2021 – Dec 2021

- Owned the initial research and development of a new resin cartridge valve for the Form 4 resin cartridge
 - Resulted in a bite valve which had a flow rate improvement of up to 280%, was easier to manufacture, and decreased the amount of resin wicked onto the valve actuator arm of the printers
- Developed a process for quickly preparing 3D printed molds for silicone casting, cutting the time taken to prepare the molds from 2 days to 1 day, allowing for a faster iterative design process while avoiding silicone cure inhibition
- Developed test programs for new technologies to improve the usability of Formlabs 3D printers, through designing and building test stands; and creating Python applications and MATLAB data analysis and visualization scripts

Engineering Design Co-op | Linamar | Guelph, ON

Sept 2018 – Dec 2018

- Engineered and implemented machine components such as jigs and pressing tools for manufacturing systems to repair and improve machines in a timely and cost-effective manner
- Worked as part of a team to optimize a new production line by running material studies, collecting and analyzing data, and presenting findings in written reports and presentations
- Conducted FEA testing to analyze and verify the failure mode of a tool which was not performing as expected

Equipment Designer – Drivetrain Group Co-op | Tigercat | Kitchener, ON

Jan 2018 – May 2018

- Improved thermal performance of a hydraulic gearbox by developing an into-mesh lubrication system; utilized 3D printed prototyping and conducted practical testing to build and correlate accurate SolidWorks CFD models
- Collaborated with an engineering team in the preliminary design stage of the TCi 920 Forestry Dozer, focusing on the suspension system as well as the tractive force and ground pressure calculations
- Increased productivity of bogie assembly line through the design of a production stand that can hold multiple sizes of bogies up to 4000 kilograms with minimal change in setup

SKILLS

CAD/FEA: SolidWorks (FEA, CFD, TolAnalyst), NX, Solid Edge, Onshape, AutoCAD, Inventor, Ansys

Prototyping: GD&T, 3D Printing, Practical Testing, Correlation Studies

Programming: Visual Basic, Python, MATLAB, Arduino, PIC